A positive chord generator from the tracial seed
Fixed-degree contraction of antisymmetric descendants

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Abstract

We connect the singlet boost realization to explicit operators in the one-matter chord Hilbert space. A grade-wise operator bound controls the nearly null antisymmetric descendants at every fixed degree. Their limit is the positive discrete series with label $d = 2$. The same bound proves positivity of a two-sided chord operator Q_h already at finite h. Its seed spectral measures converge to the gamma law of the preceding translation construction. This is a model-level construction with the correct microscopic boost; selecting physical momentum units and the interacting multiparticle dynamics remains the next step.

1 The construction and its microscopic clock

Use the full annealed trace at $\beta = \mu_{\rm chem} = 0$, first the complex DSSYK double scaling limit at fixed $\lambda = p^2/N$, then $h = 4\lambda \to 0$. The conserved-subtracted derivative singlet of our previous notes has matter weight q^d, with $q = e^{-h}$ and $d = 2$. We work in its one-matter chord representation, not in a finite-N sample Hilbert space. The proof below applies to every fixed $d > 0$. Let $\mathcal{H}_n$ be the grade-n space and $\Omega = |d; 0, 0\rangle$, with $\|\Omega\| = 1$. Define

$$E_h = \bar{n}, \quad E_h|_{\mathcal{H}_n} = d + n, \quad C = a_L^\dagger - a_R^\dagger, \quad D = C^\dagger = a_L - a_R,$$
$$K_+ = C/\sqrt{h}, \quad K_- = D/\sqrt{h}, \qquad B_h = \frac{K_+ + K_-}{2}, \quad \Pi_h = \frac{K_- - K_+}{2i}, \quad \boxed{Q_h = E_h + \Pi_h.} \tag{1}$$

With $H_{\rm src} = H_s/(2\mathcal{J}_s) = (a^\dagger + a)/\sqrt{h}$,

$$B_h = \frac{H_{{\rm src},L} - H_{{\rm src},R}}{2} = \frac{H_{s,L} - H_{s,R}}{4\mathcal{J}_s}, \qquad \eta = 4\mathcal{J}_s t. \tag{2}$$

Thus B_h is the actual singlet Liouvillian in boost units. The main claims proved below are $Q_h \geq 0$ for every $h > 0$, and convergence on each finite descendant span to $[B, Q] = iQ$ with $Q = p$ on $L^2(\mathbb{R}_+, dp)$. Here p, Q, E, B are dimensionless.

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2 A bound that survives the degenerate Gram limit

The starting identities are the chord algebra of Lin–Stanford [1], Eqs. (22)–(25):

$$[a_j, a_j^\dagger]_q = 1, \quad [a_L, a_R^\dagger] = [a_R, a_L^\dagger] = q^{E_h},$$
$$[E_h, a_j^\dagger] = a_j^\dagger, \quad a_j\Omega = 0. \tag{3}$$

All norms below use the physical chord Gram inner product. Set $\delta = 1-q$, $[n]_q = (1-q^n)/(1-q)$ and $A = a_L^\dagger a_L + a_R^\dagger a_R \geq 0$. Direct subtraction gives

$$[D, C] = 2(1-q^{E_h}) - \delta A. \tag{4}$$

First, $\|a_j|_{\mathcal{H}_n}\|^2 \leq [n]_q \leq n$. Indeed the adjoint block and $a_j a_j^\dagger = 1 + q a_j^\dagger a_j$ give the induction $s_{n+1} \leq 1 + q s_n$, starting at $s_0 = 0$. The same argument makes a_j bounded on the full direct sum, with $\|a_j\| \leq \delta^{-1/2}$ at fixed $h > 0$.

Let $t_n = \|C : \mathcal{H}_n \to \mathcal{H}_{n+1}\|^2$ and $t_{-1} = 0$. Since $D|_{\mathcal{H}_n}$ is the adjoint of the preceding creation block, Eq. (4) implies

$$\|Cf\|^2 = \|Df\|^2 + 2(1-q^{d+n})\|f\|^2 - \delta\langle f, Af\rangle,$$
$$t_n \leq t_{n-1} + 2(1-q^{d+n}) \leq 2\sum_{j=0}^{n}(1-q^{d+j}) \leq h(n+1)(2d+n). \tag{5}$$

Consequently $K_\pm$ are uniformly bounded between any fixed pair of grades as $h \to 0$, including directions whose coordinate Gram matrices become singular. This is the step a coefficient-wise Taylor expansion alone cannot supply.

Put $w_n = K_+^n\Omega = h^{-n/2}C^n\Omega$. Equation (5) gives $\|w_n\|^2 \leq n!(2d)_n$. Moreover,

$$[a_L, K_+] = \frac{1 - q^{E_h} - \delta a_L^\dagger a_L}{\sqrt{h}}, \quad \|[a_L, K_+]|_{\mathcal{H}_k}\| \leq (d+2k)\sqrt{h}. \tag{6}$$

The a_R commutator has the opposite sign with L replaced by R. Commuting either annihilator through the finite product K_+^n, using Eq. (5) at each step, proves

$$\|a_L w_n\| + \|a_R w_n\| = O_{n,d}(\sqrt{h}), \qquad \|Aw_n\| = O_{n,d}(\sqrt{h}). \tag{7}$$

The last estimate also uses the bounded creation blocks on fixed grades. These are norm estimates, not cancellations inferred from coordinates.

3 The fixed-degree discrete-series limit

From Eqs. (4) and (7),

$$([K_-, K_+] - 2(d+n))w_n = O_{n,d}(\sqrt{h}) \quad \text{in Hilbert norm.} \tag{8}$$

Here $2(1-e^{-h(d+n)})/h = 2(d+n) + O(h)$ and $\delta/h \le 1$. Expand $[K_-, K_+^n]$ into n terms. Every error remains small under the remaining creation blocks by Eq. (5), giving

$$K_- w_n = n(2d+n-1)w_{n-1} + O_{n,d}(\sqrt{h}). \tag{9}$$

Taking its inner product with w_{n-1} proves inductively

$$\|w_n\|^2 = n!(2d)_n + O_{n,d}(\sqrt{h}), \qquad \lim_{h\to 0} h^{-n}\|C^n\Omega\|^2 = n!(2d)_n. \tag{10}$$

Different grades are orthogonal. Hence $e_n(h) = w_n/\|w_n\|$ defines an orthonormal family and, for every fixed n,

$$\begin{aligned} K_+ e_n(h) &= \sqrt{(n+1)(2d+n)}\, e_{n+1}(h) + O(\sqrt{h}), \\ K_- e_n(h) &= \sqrt{n(2d+n-1)}\, e_{n-1}(h) + O(\sqrt{h}), \\ E_h e_n(h) &= (d+n)e_n(h). \end{aligned} \tag{11}$$

The lowering error can have components outside this family at finite h. We have controlled those components in the full Hilbert norm. Any fixed polynomial word in $K_\pm, E_h$ therefore converges on a fixed finite span. The limiting operators obey

$$[K_-, K_+] = 2E, \quad [E, K_\pm] = \pm K_\pm, \qquad E^2 - B^2 - \Pi^2 = d(d-1). \tag{12}$$

In particular the boost Jacobi coefficients are $b_n = \frac{1}{2}\sqrt{n(2d+n-1)}$.

Relation to the existing Krylov construction. The signed binomial vectors $C^n\Omega$ are already the vectors of Ambrosini et al. [2], Eq. (81), up to orientation. Their exact lowering coefficients, Eq. (91), are

$$c_k(n) = n\frac{[n-k]_q}{n-k}(1-q^{d+k}) + n\frac{[k+1]_q}{k+1}(1-q^{d+n-1-k}). \tag{13}$$

They expand as $hn(2d+n-1) + O(h^2)$ at fixed n, d. Their finite-q binomial Krylov ansatz develops tails from level four. Their fixed-$\widetilde{q}$ heavy limit and fixed-hn semiclassical limit differ from ours. We use their established vectors and give the norm control for the fixed-d, fixed-degree application here; we do not claim discovery of the binomial basis or the discrete series.

4 Positive Q_h and its explicit limiting representation

Define the normalized real Laguerre functions and their phased version

$$g_n(p) = \sqrt{\frac{n!}{(2d)_n}}\,\frac{2^d p^{d-1/2} e^{-p}}{\sqrt{\Gamma(2d)}} L_n^{(2d-1)}(2p), \qquad \phi_n = i^n g_n. \tag{14}$$

They form a complete orthonormal basis of $L^2(\mathbb{R}_+, dp)$. The three-term Laguerre recurrence and its derivative identity give

$$E\phi_n = (d+n)\phi_n, \qquad B = i(p\partial_p + \tfrac{1}{2}), \qquad Q = E + \Pi = p. \tag{15}$$

The matrix elements agree with Eq. (11), with $\Pi = (K_- - K_+)/(2i)$ and the phases fixed by Eq. (14). The Jacobi off-diagonals grow linearly, so their reciprocal sum diverges. Carleman's criterion gives essential self-adjointness of B, Q on the finite basis span; Q closes to positive multiplication by p. For B, logarithmic coordinates give the usual translation generator. This also fixes the domains of the limiting observables.

Finite-h positivity. Write $f = \sum_n f_n$ with finitely many nonzero grades and set $r_n = \|f_n\|$. For $c_n = \sqrt{(n+1)(2d+n)}$, Eq. (5) implies

$$\langle f, Q_h f\rangle \geq \sum_n (d+n) r_n^2 - \sum_n c_n r_n r_{n+1} = \int_0^\infty p \left|\sum_n r_n g_n(p)\right|^2 dp \geq 0. \tag{16}$$

At fixed h, Π_h is bounded and symmetric, whereas E_h is self-adjoint on the direct-sum number domain. Thus $Q_h = E_h + \Pi_h$ is self-adjoint on $D(E_h)$, finite-grade vectors are a core, and Eq. (16) proves $Q_h \geq 0$ on the full one-matter space.

The exact covariance defect is

$$[B_h, Q_h] - iQ_h = i\left\{\frac{1 - q^{E_h}}{h} - E_h - \frac{\delta}{2h}A\right\}. \tag{17}$$

It tends to zero on fixed descendants by Eq. (7). Positivity is exact at finite h; dilation covariance is a limiting statement on the specified domain. Q_h is a two-sided chord operator, so this does not contradict the previous obstruction for a positive one-sided observable with exact trace-preserving dilation.

5 A spectral prediction and numerical checks

The seed is $\phi_0 = \psi_d$. Its limiting Q measure and characteristic function are explicit:

$$d\nu(p) = \frac{2^{2d}}{\Gamma(2d)} p^{2d-1} e^{-2p} dp, \quad \langle Q^k \rangle = \frac{(2d)_k}{2^k}, \quad \langle e^{-isQ} \rangle = (1 + is/2)^{-2d}. \tag{18}$$

Polynomial-word convergence proves convergence of every seed moment of Q_h. Positivity and the bounded first moment give tightness; bounded higher moments give uniform integrability for each fixed moment. The gamma law is moment-determinate (its moment generating function is finite near zero). Hence the entire seed spectral measure converges weakly to Eq. (18), including its characteristic function at every fixed real s. An exact finite-h check is

$$\langle Q_h \rangle = d, \qquad \mathrm{Var}_\Omega(Q_h) = \frac{1 - q^d}{2h} \longrightarrow \frac{d}{2}. \tag{19}$$

The limiting boost gives $\langle e^{-i\eta B} \rangle = \mathrm{sech}^{2d}(\eta/2)$. Within this limiting representation its descendant probabilities are the standard discrete-series result

$$p_n(\eta) = \frac{(2d)_n}{n!} \tanh^{2n}(\eta/2) \, \mathrm{sech}^{4d}(\eta/2), \quad \sum_n n p_n = 2d \sinh^2(\eta/2). \tag{20}$$

This last formula describes the limiting representation; no uniform finite-h, growing-time chord-number theorem is asserted.

Checks saved with this note. Exact Gram polynomials at $d = 2$, $1 \le n \le 6$ give leading norms $4, 40, 720, 20160, 806400, 43545600$ multiplying $(1-q)^n$. Their normalized lowering residuals are $O(h)$ at these tested degrees, stronger than the sufficient $O(\sqrt{h})$ bound above. Independent recursive Gram computations at $d = 0.7, 2$ and $h = 0.1, 0.02, 0.005$ check whole-grade creation bounds ($n \le 4$), descendant norms ($n \le 6$), and Q_h moments ($k \le 8$), using 80 and 110 decimal digits. Eighteen exact symbolic identities check the B, Q Laguerre actions through $n = 8$.

h	$\|w_6\|^2/[6!(4)_6]$	relative error in $\langle Q_h^8 \rangle$
0.1	0.2705804	0.216685
0.02	0.7646896	0.049444
0.005	0.9348279	0.012686

The largest relative change on precision refinement is below 2.1×10^{-69}. This is an empirical arithmetic check, not interval certification. Finite-h discrepancies in the table are physical approximation errors, not numerical rounding errors.

6 What this adds to the Susskind research route

The previous stage built $Q = p$ as an extension of the boost kernel. We now have an explicit positive chord representative Q_h, a controlled descendant limit, and convergence of its seed spectral law. Thus the candidate is anchored to the correct microscopic Liouvillian and number operator, rather than only to a scalar two-point fit. Equations (5)–(16) are the argument developed in this note from the cited chord algebra. Priority for these consequences has not been established by an exhaustive literature search.

In the limiting one-particle representation one can again set

$$P^+ = \mu Q, \quad P^- = \frac{m^2}{\mu Q}, \quad P^{0,1} = \tfrac{1}{2}(P^+ \pm P^-), \qquad m, \mu > 0. \tag{21}$$

Both m and μ have energy units. This is the preceding standard positive-energy mass-family construction, not a mass prediction. The existence of Q_h does not establish convergence of its inverse: weak spectral convergence does not control the singular endpoint $p = 0$. Nor does the one-matter construction specify the interacting two-matter P^- or a meson spectrum.

The next calculation is therefore the action of this construction on two-matter states, using the actual Lin–Stanford coproduct and Gram inner product. Their creation maps are $D(a_L^\dagger) = a_L^\dagger \otimes 1$ and $D(a_R^\dagger) = 1 \otimes a_R^\dagger$; annihilation maps contain $q^{\bar{n}}$ factors. A primitive sum of independently postulated particle momenta cannot simply be substituted. We should determine which combinations survive on relative descendants, then confront them with the already derived three-channel singlet four-point function. This is where a kinematic construction can begin to test interaction and mass data.

Reproducibility and scope. The exact exploratory output, independent validation, source mapping, and input/output SHA-256 hashes are saved under the `susskind-antisymmetric` stage. The manuscript is built by `code/build_antisymmetric_chords.ps1`, using two `pdflatex` passes. The source and numerical records remain local; no Git commit, push, or archive was made in this stage.

References

[1] H. W. Lin and D. Stanford, *A symmetry algebra in double-scaled SYK*, arXiv:2307.15725. Used: Secs. 2.2, 2.3 and 3.2, chord algebra, coproduct and one-matter action.

[2] M. Ambrosini, E. Rabinovici, A. Sánchez-Garrido, R. Shir and J. Sonner, *Operator K-complexity in DSSYK: Krylov complexity equals bulk length*, arXiv:2412.15318v2. Used: Secs. 3.1, 3.2 and 3.4, especially Eqs. (81), (90)–(95).